BUS BOOKING MANAGEMENT SYSTEM — FULL SYSTEM ARCHITECTURE

Role-based web application • React frontend • Node.js / Express backend • MongoDB Atlas • External verification and payment services

1. PRESENTATION LAYER

Customer UI
Search buses • Select seats
Book / cancel tickets • View PNR

Operator UI
Manage buses • Routes • Trips
Schedules • Passenger manifest

Admin UI
Manage users and operators
Monitor bookings • Reports

FRONTEND TECHNOLOGY
React.js + Vite + Tailwind CSS + React Router • Role-based navigation

HTTPS / REST / JSON
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2. APPLICATION LAYER

NODE.JS + EXPRESS.JS REST API
HTTPS endpoints • Routing • Middleware • Request validation • Centralized error handling

AUTHENTICATION & RBAC
Registration • Login • JWT
Email verification

USER & OPERATOR MANAGEMENT
Profiles • Roles • Operator accounts
Account status and permissions
Operator Registration , admin Approval

Email verification

Read trip availability

BUS & ROUTE MANAGEMENT
Bus details • Capacity • Stops
Source / destination / route

TRIP & SCHEDULE MANAGEMENT
Travel date • Departure / arrival
Fare • Trip availability

SEAT AVAILABILITY & LOCKING
Seat map • Availability checks
Temporary lock • Expiry / release

Reserve / confirm seats

BOOKING & PNR MANAGEMENT
Passenger details • Booking status
Unique PNR • E-ticket

PAYMENT & CANCELLATION
Payment Processing & Simulated Refund Status
Cancellation Policy • Cancellation Request • Refund Status.

Booking / cancellation email

REPORTS & AUDIT LOGS
Booking / occupancy reports
Admin activity and event history

Payment / refund API

Mongoose ODM • Models • Business rules • Database transactions / atomic updates

Atomic lock / release
Persist booking / PNR
Mongoose operations

3. DATA LAYER

MongoDB Atlas
Persistent document database

- Users / Operators
- Buses / Routes
- Trips / Schedules
- Seat Inventory
- Bookings / Passengers
- Payments / Refunds
- Notifications
- Audit Logs

4. EXTERNAL SERVICES

EMAIL SERVICE
Nodemailer / Gmail SMTP
Verification links or OTP • Booking email

Verified webhook / result

PAYMENT GATEWAY
Gateway test mode
Payment processing • Refund API

5. SECURITY & QUALITY — CROSS-CUTTING CONCERNS

- JWT authentication
  Role-based access control
- bcrypt password hashing
  Secure secrets / HTTPS
- Email
  Expiry • Rate limiting
- Atomic seat locking
  Hold expiry • Idempotency
- Input validation
  Error handling • Testing
- Audit logging
  Backup • Monitoring

LEGEND Blue = Presentation Green = Application Purple = Database Grey = External services Yellow = Security / Quality Solid arrows = API / data flow Dashed arrows = External integration / callback